



Enterprise AI-Powered Compliance & Governance Platform

Automated Document Auditing and Multi-Agent Consensus

LANGGRAPH

AWS BEDROCK

POSTGRESQL / SQLITE

FAISS

STREAMLIT

Automating PII detection, confidentiality screening, threat validation, and compliance reporting at enterprise scale.

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The Enterprise Risk & Cost Dilemma



Manual Processing Bottlenecks

High-volume contract and document auditing introduces human error and critical security oversights.

Data Privacy Regulations

GDPR, CCPA, and SOC2 penalize unauthorized PII exposure and confidentiality leaks.

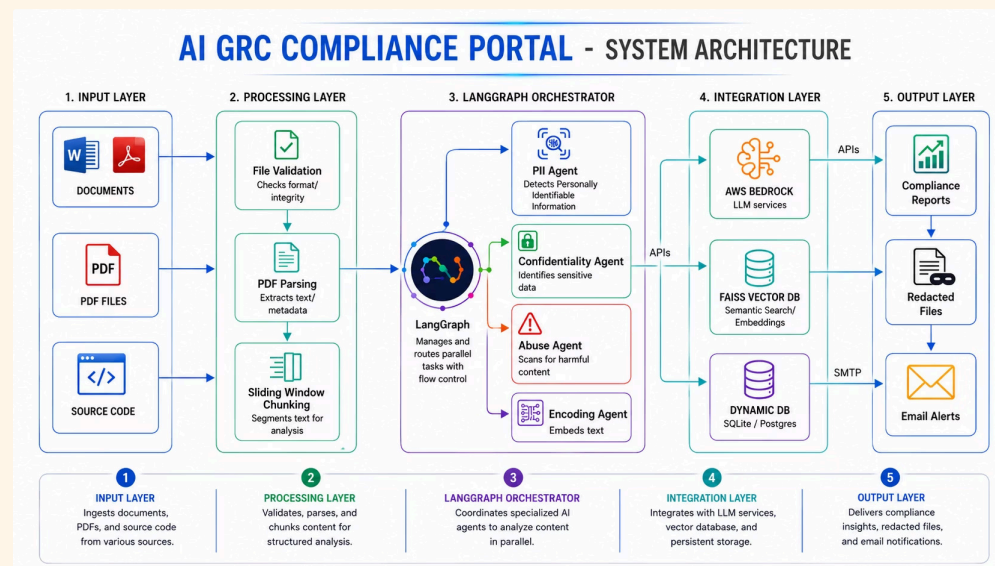
High Cost of LLM Auditing

Standard LLM verification pipelines consume excessive tokens, driving up costs for repetitive scanning.

System Brittleness

Tight coupling to a single database or LLM provider halts workflows when environments change.

An Intelligent, Self-Optimizing GRC Engine



MoE Agents

Parallel specialist LLM agents analyze documents simultaneously across custom domains.

Zero-Trust Scanners

Prompts and files are vetted for injections and format violations before processing.

Hybrid Architecture

Local SQLite for testing, seamless AWS Bedrock + PostgreSQL for production.

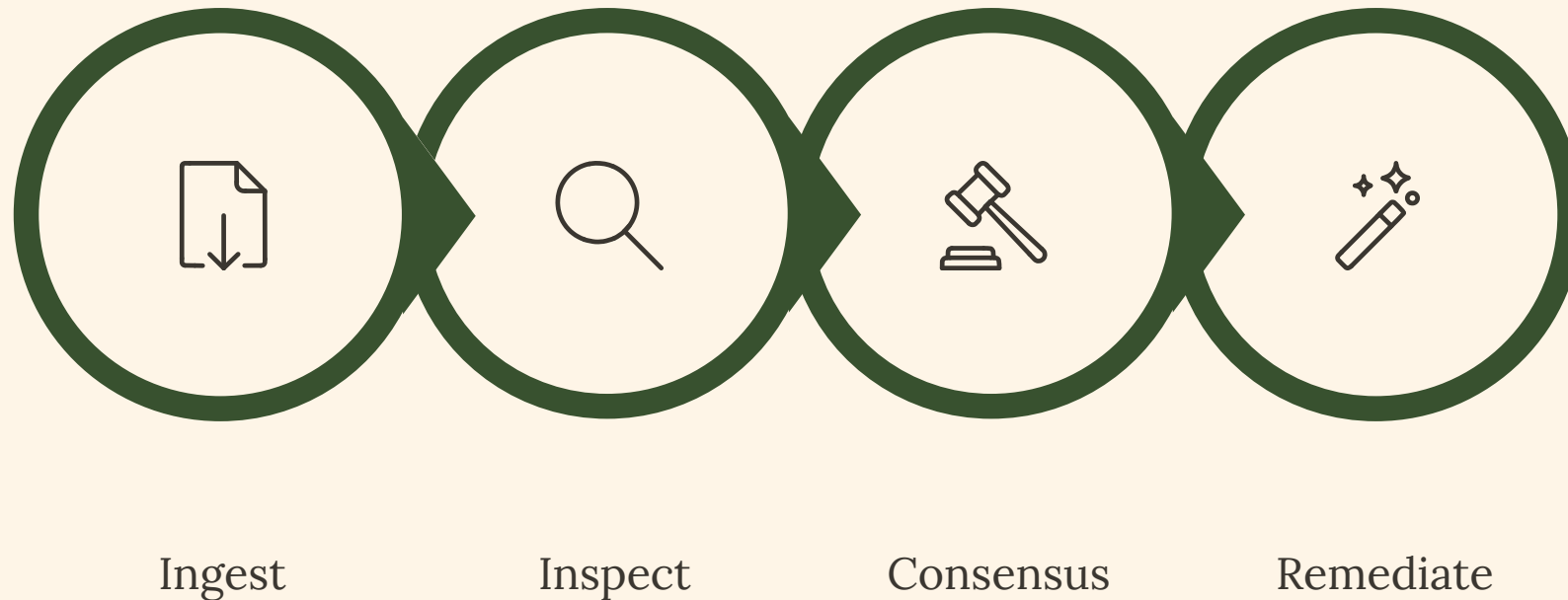
Actionable Outputs

Auto-generated redacted PDFs, consensus ledgers, and real-time violation alerts.

LangGraph Orchestrated Compliance Pipeline



LangGraph's graph-based orchestration enables parallel agent execution with deterministic consensus merging — minimizing latency while maintaining audit traceability across every document processed.



Data Ingestion to In-Depth Issue Visibility



Our platform transforms raw data into actionable intelligence, providing a comprehensive and granular view of compliance issues.



Universal Extractor

Safely processes diverse formats like .pdf, .py, .ts, .go, .sh, .yaml, .csv, ensuring robust handling with UTF-8 error replacement.



Simulated Chunking

Breaks plain-text and source files into 3,000-character pages, preserving file structure for efficient processing by MoE agents.



Zero-Trust Guardrails

Ingestion-time checks instantly block potential prompt injections and reject files exceeding 10MB to maintain system integrity.



Categorized Violations Console

Provides a real-time, high-fidelity view of findings, categorized by PII, Confidentiality, Threats, and Encoding. Includes line numbers, code snippets, severity tags (Low-Critical), and policy context.

The Mixture-of-Experts (MoE) Agent Network



Five specialist agents collaborate in parallel, each owning a distinct compliance domain. A Reviewer agent acts as the panel judge, consolidating findings into a unified risk report.



PII Agent

Scans for emails, phone numbers, API keys, and credit cards. Regex pre-screening bypasses the LLM when clean.



Confidentiality Agent

Screens for financial forecasts, source code, pricing models, and trade secrets.



Abuse Agent

Flags threats, harassment, hate speech, and unlawful content.



Encoding Agent

Checks for Mojibake, invalid character sets, and text corruption programmatically.



Reviewer Agent

Standardizes alerts, deduplicates conflicts, and ranks severity from LOW to CRITICAL.

Smart Policy Auditing & FAISS Integration



Timestamp	Event Type	Target Resource	Log Details
2026-06-04 06:17:34	REPORT_GENERATED	supervisor_certificate.pdf	Audit report created. Score: 93.0
2026-06-04 06:17:31	FILE_SYNCED	manual_1780553841_supervisor_certificate.pdf	Successfully synced metadata from S3 bucket 'compliance-governance-bucket'.
2026-06-04 06:17:29	FILE_DETECTED	manual_1780553841_supervisor_certificate.pdf	New file detected in AWS S3: manual_1780553841_supervisor_certificate.pdf
2026-06-04 06:17:23	FILE_ANALYZED	supervisor_certificate.pdf	Manual scan execution initiated. Scan ID: 14
2026-06-03 10:50:47	REPORT_GENERATED	violation_pii_leak.pdf	Manual Approval Scan Complete. Score: 70.0/100. Report saved.
2026-06-03 10:50:34	FILE_ANALYZED	violation_pii_leak.pdf	Manual Approval Analysis initiated. Scan ID: 13
2026-06-03 10:50:31	REPORT_GENERATED	violation_confidentiality_leak.pdf	Manual Approval Scan Complete. Score: 70.0/100. Report saved.
2026-06-03 10:50:19	FILE_ANALYZED	violation_confidentiality_leak.pdf	Manual Approval Analysis initiated. Scan ID: 12
2026-06-03 10:50:16	REPORT_GENERATED	violation_abuse_threats.pdf	Manual Approval Scan Complete. Score: 100.0/100. Report saved.
2026-06-03 10:50:10	FILE_ANALYZED	violation_abuse_threats.pdf	Manual Approval Analysis initiated. Scan ID: 11
2026-06-03 10:50:07	REPORT_GENERATED	needle_in_haystack_performance.pdf	Manual Approval Scan Complete. Score: 70.0/100. Report saved.
2026-06-03 10:49:53	FILE_ANALYZED	needle_in_haystack_performance.pdf	Manual Approval Analysis initiated. Scan ID: 10
2026-06-03 10:49:50	REPORT_GENERATED	mojibake_encoding_trap.pdf	Manual Approval Scan Complete. Score: 98.0/100. Report saved.
2026-06-03 10:49:40	FILE_ANALYZED	mojibake_encoding_trap.pdf	Manual Approval Analysis initiated. Scan ID: 9
2026-06-03 10:49:37	REPORT_GENERATED	manual_1780479207_semantic_threat_trap.pdf	Manual Approval Scan Complete. Score: 40.0/100. Report saved.
2026-06-03 10:49:22	FILE_ANALYZED	manual_1780479207_semantic_threat_trap.pdf	Manual Approval Analysis initiated. Scan ID: 8
2026-06-03 10:49:19	REPORT_GENERATED	large_paginated_text.txt	Manual Approval Scan Complete. Score: 10.0/100. Report saved.
2026-06-03 10:48:58	FILE_ANALYZED	large_paginated_text.txt	Manual Approval Analysis initiated. Scan ID: 7

FAISS Vector Store

Matches detected violations directly to relevant policy clauses using semantic similarity search.

Titan Embeddings

Uses amazon.titan-embed-text-v2:0 to vectorize corporate guideline books for precise retrieval.

Verifiable Justifications

The RAG node forces the LLM to cite policy context, drastically reducing false positives.

Cost Controls & Runtime Analytics



Regex Bypass Engine

PII scans use rapid regex filters first. If no patterns match, the LLM call is completely bypassed — reducing token consumption by up to **80%**.

AWS Bedrock Pricing Adaptability

Fine-tuned for Amazon Nova models: **nova-micro** for fast checks and **nova-lite** for complex RAG tasks — offering competitive per-token pricing.

Real-Time Analytics Dashboard

Visual dashboards chart token usage, latency, and estimated cost per model run in real time.

NOVA LITE CALLS

57

NOVA MICRO CALLS

54

AVG LATENCY

0.65s

ESTIMATED COST

\$0.0057

Automated Incident Response & Auditing



PDF COMPLIANCE AUDIT REPORT

Target Document: tmp797en8z6.py
Generated On: 2026-06-03 10:48:50

COMPLIANCE SCORE	OVERALL RISK RATING	TOTAL VIOLATIONS
55.0/100	HIGH	2

Executive Summary

The document exhibits a HIGH risk level with a compliance score of 55.0/100, primarily due to critical confidentiality and abuse policy violations. The detected issues include the exposure of highly sensitive AWS access keys and a comment suggesting malicious intent to blackmail developers. Immediate remediation is required to remove or secure sensitive credentials and eliminate threatening content to mitigate these risks.

Detected Policy Violations Ledger

Page	Category	Severity	Sensitive Snippet	Policy Clause / AI Reasoning
1	Confidential	CRITICAL	AWS_ACCESS_KEY_ID = "AKIAUJKSHYTDQERJ99AA" "AWS_SECRET_ACC..."	Policy Link: tmpjcca00a.pdf (Page 1) The text contains AWS access keys, which are highly sensitive credentials that provide full access to the AWS account. This is considered a critical security risk as it can lead to unauthorized access and potential data breaches. Policy Match: The snippet contains a comment that includes a threat to blackmail developers if a build fails, which violates the zero-tolerance policy for hostile communications. The policy explicitly prohibits any text containing threats to blackmail personnel or execute coercive actions against external teams.
1	Abuse	HIGH	"Let's threat and blackmail their developers if the build..."	Policy Link: tmpjcca00a.pdf (Page 1) The text suggests a malicious intent to threaten and blackmail developers, which is a clear violation of legal and ethical standards. Policy Match: The snippet contains a threat to blackmail developers if a build fails, which violates the zero-tolerance policy for hostile communications. The policy explicitly prohibits any text containing threats to blackmail personnel or execute coercive actions against external teams.

CRITICAL COMPLIANCE VIOLATION: violation_pii_leak.pdf

J

jayantchawla29@gmail.com
to me

Warning: A GRC compliance audit has flagged critical violations in document: 'violation_pii_leak.pdf'.

Scan Details:

- Final Compliance Score: 70.0/100
- Total Violations Found: 1
- System Assessment: CRITICAL RISK DETECTED

Please review the logs immediately in the Case Overrides console.

Reply

Forward

Every scan produces a defensible, audit-ready output package — combining professional PDF reports, coordinate-based redaction, and automated team alerts.

ReportLab PDF Generator

Compiles multi-page audit reports with incident descriptions, risk scorecards, and metadata.

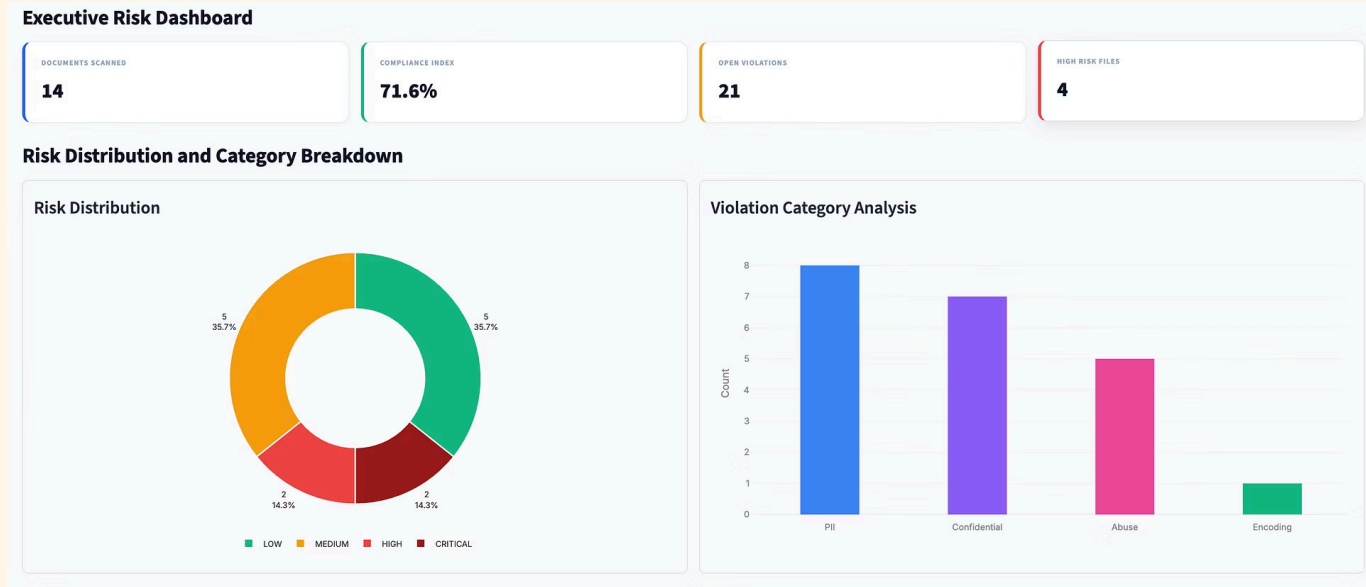
Smart Coordinate Redaction

Coordinate-based PDF redaction scrubs high-risk content directly from source documents.

SMTP Email Alerting

Triggers alerts to security teams for CRITICAL-risk files or infrastructure failures, with mock fallbacks for safe testing.

Interactive Governance Console



Our intuitive dashboard provides a centralized hub for managing compliance, testing policies, and monitoring system health with precision.

Case Overrides Console

Empowers compliance officers to review and override false-positive alerts, manually adjust risk scores, and save updated audit states, ensuring granular control over the compliance workflow.

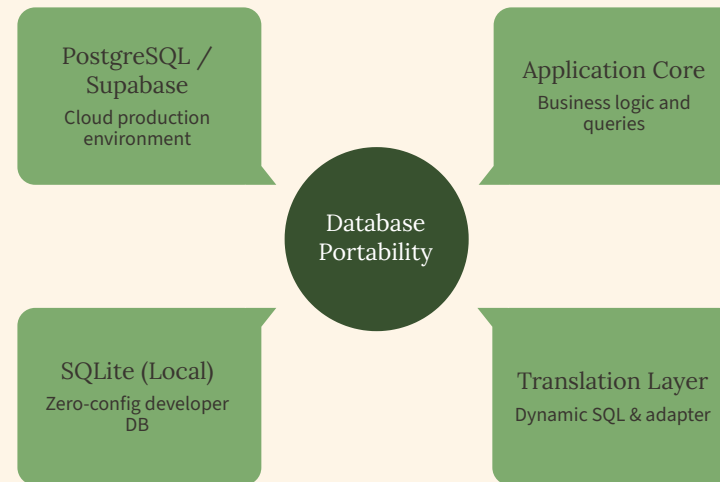
Regex Sandbox

An interactive environment for live testing and validating custom regex patterns against sample data before deploying them into the production scanning pipeline, minimizing errors and optimizing performance.

System Diagnostics

Provides real-time monitoring of critical infrastructure metrics, including database connection pools, environment variable integrity, thread pool utilization, and S3 bucket status.

Zero-Code Change Database Portability



Transparent SQL Parser & Interceptor

An active SQL decorator layer intercepts raw queries, enabling a single query pattern across diverse SQL engines.



Dialect Mapping & Parameter Substitution

Dynamically translates `?` to `%s` parameters and converts `INSERT OR IGNORE` to ANSI SQL `ON CONFLICT` statements.



ORM Compatibility Emulation

Automatically injects `RETURNING id` on PostgreSQL inserts, preserving `cursor.lastrowid` compatibility for ORMs.



Configuration-Driven Toggle

Retargets the system DB from local to cloud via `DATABASE_URL` environment variable changes, ensuring instant re-platforming.

Scaling AI Governance for the Future



Our future roadmap focuses on continuously enhancing our AI governance platform, ensuring adaptability and resilience in an evolving regulatory landscape.

Real-Time Event Stream Scanning

Integrating with Apache Kafka or AWS Kinesis to scan documents as they are uploaded, ensuring immediate compliance checks.

Interactive UI Redaction Integration

Activating the backend PyMuPDF redaction pipeline directly within the Streamlit Case Management panel to enable one-click document sanitization.

Automated Guardrails

Implementing AWS Bedrock Guardrails to systematically block malicious prompt injections and maintain system integrity.

Resilient & Self-Optimizing

A platform designed for continuous learning, combining advanced agent logic, hybrid databases, and strict cost controls for unparalleled future-proof compliance.



Thank You

We appreciate your time.